

Pranjal T S

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Education

California State University, East Bay
MS, Statistics(Data Science Concentration)

Aug 2025 - Jun 2027
Hayward, CA

PES University
Bachelor of Technology, Computer Science

Sep 2021 - May 2025
Bengaluru, India

Professional Experience

Toyota Kirloskar Motor Private Limited
Applied Machine Learning Intern

Jan 2025 - Jun 2025
Bangalore, India

- Developed an Automated Incident Response Metric System, an incident-analysis system that processed 10,000+ real security events, applied token-based classification, and generated SLA metrics to support priority-based response.
- Identified 31% of P2 incidents breaching SLA during night shifts, surfacing a critical staffing gap and enabling targeted resource reallocation across security operations.
- Replaced manual monthly reporting process by automating incident analysis workflows, cutting report generation time by 70% and saving 15+ analyst hours per month.

Technical Skills

- **Languages:** Python, R, SQL
- **Libraries:** pandas, Numpy, PyTorch, Scikit-learn, Hugging Face Transformers, Keras, TensorFlow, Matplotlib, OpenCV, Seaborn
- **Machine learning:** Classification, Regression, Clustering, CNNs, Transformers, RAG, NLP, Computer Vision, Feature Engineering, Model Evaluation
- **Cloud & Tools:** Docker, Git, Jupyter, Google Colab, AWS, Weights & Biases, Hugging Face Hub

Key Projects

Customer churn prediction [Website Link](#)

Feb 2026 - Apr 2026

- Ranked telecom customers by churn and revenue risk to prioritise outreach, identifying \$365K in at-risk revenue.
- Fixed blank billing charges and inconsistent service categories into 33 features, then tuned decisions to catch 58% of churners.
- Selected Logistic Regression as the final churn model, ranking churners above non-churners 80% of the time.

RAG based Healthcare Chatbot [Website Link](#)

Nov 2025 - Jan 2026

- Reduced hallucination in medical answers from 27% in general LLM's to 6% by forcing the chatbot to only use verified sources.
- Transformed unstructured data from NIH PubMed and medical encyclopedias into 1,588 clean passages, achieving 92% answer accuracy.
- Added a reranking stage that improved answer faithfulness from 80% to 91.7%, generating verified medical responses in under 1.4 seconds.

Production Defect Detection System [Website Link](#)

Oct 2025 - Dec 2025

- Built automated system, reducing manual bottle surface inspection from 5 seconds to 0.04 seconds per unit.
- DenseNet alone wasn't reliable on production bottles; built dual-model system with Autoencoder for cross-verification.
- Tuned parameters in each model, which improved defect detection precision from 50% to 93% on validation data.

LLM Fine-Tuning Optimizer

Feb 2025 - Jul 2025

- Eliminated manual LLM fine-tuning testing with a self-optimizing search system, cutting computational cost by 98% from 87.5M to 1.2M parameters.
- Engineered blockwise crossover and layerwise mutation operators using NSGA-II to optimize LoRA configurations.
- Resolved parameter corruption across 5 BERT family models by replacing random parameter swapping with whole-layer exchange, achieving 93.80% accuracy across 3 NLP tasks.

Publications

- Pranjal T.S et al..Healthy Life Expectancy (HALE) Analysis and Prediction using Machine Learning.2025 *International Conference, IEEE*, 2025. <https://ieeexplore.ieee.org/document/10987496>